Local-KLE (multiscale) pCN samplers on a Darcy inverse problem: pre-sampling tests and production results

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Abstract

We compare six pCN samplers for the log-permeability coefficients of a 32×32 Darcy problem (no β, no ρ; squared-exponential prior, $\ell = 0.1$, $r = 64$ modes): global pCN (1), local-KLE primal raw (2), primal whitened with a pseudo-inverse (3a) and with the null-space term (3b), a primal/dual mixture (4), and a primal→dual sequence (5). Whitening is exact only for the full-rank case; the primal 4×4 decomposition with 4 local modes per block has rank $\Sigma_{ms} = 58 < 64$. Prior tests (likelihood off) and production runs (64 chains, 10^5 iterations) agree on the ordering of the samplers. **3b is statistically indistinguishable from global pCN; 3a, 2 and 5 show a modest loss of posterior spread (about 10–17%); experiment 4 fails in practice** because its primal move is almost never accepted (0.16%), and its results should not be read as a posterior.

1 Summary of findings

1. **Rank.** The primal $\Sigma_{ms} = BB^\top$ is rank deficient by exactly six directions ($r_+ = 58$); the dual one is full rank ($r_+ = 64$, cond 10.7). Whitening therefore needs either the pseudo-inverse (3a) or the report's null-space term (3b).

2. **3b reproduces global pCN.** Prior test: covariance error 0.023 for both. Production: mean posterior sd of $\log \kappa$ 0.575 vs. 0.570, same acceptance, same relative-error plateau, MPSRF 1.12 vs. 1.13. It is a correctness device, as predicted, not a speed-up.

3. **Raw and pseudo-inverse samplers lose spread.** Mean posterior sd of $\log \kappa$ relative to experiment 1: 2: 0.83, 3a: 0.90, 5: 0.86. The ordering matches the prior-test variance deficits (pointwise variance ratios 0.78, 0.93, 0.78/0.75).

4. **Experiment 4 is not usable as configured.** The primal move's step size sank to its floor (10^{-3}) and its acceptance is 0.16%; the chains plateau at relerr = 0.033 (others 0.020), MPSRF is 1.61, and the posterior sd collapses to 0.22 (0.38× experiment 1).

5. **All well-behaved samplers plateau at the same relative error,** relerr ≈ 0.0204**,** about 6.4× the data noise level (0.0032), as anticipated by the group; $\Phi \approx 23$ there.

6. **Experiment 5** switches from primal to dual with essentially no cost: median Φ 23.6 → 24.8, relerr 0.0206 → 0.0211.

All conclusions rest on one synthetic data set (truth seed 2026) and one set of chain seeds; no repeated-data uncertainty is quantified (Section 6).

2 Setup

Problem. Unit square, 32×32 cells ($N = 1024$); $\log \kappa = T\theta$, $\theta \sim N(0, I_{64})$ (KLE at energy 0.966 gives $r = 64$, modes 63/64 a tied pair); pressure 1 on the left and 0 on the right face, no flow top and bottom (Dirichlet permeameter, harmonic transmissibilities). Data: pressure on 512 checkerboard cells, truth a fixed-seed prior draw, additive Gaussian noise with global relative size 0.003. Likelihood $\Phi = \|d - Op\|^2/(2\sigma_{rel}^2\|d\|^2)$ with $\sigma_{rel} = 0.003$. With this global-norm likelihood the expected Φ at the truth is ≈ 0.5 (realised 0.567, relerr = 0.0032).

Samplers. All use pCN, $s_{\text{new}} = \sqrt{1 - s^2}\, x + s\xi$, acceptance on the likelihood ratio only, one forward solve per iteration, one step size per grid (all blocks proposed together), adapted to 25% acceptance for the first $M = 10^4$ iterations (each phase for 5) and then frozen. B is the assemble-and-project matrix from local KLE coefficients c to the global KLE.

	state	θ
1	θ	θ (global pCN)
2	c (primal, $p = 64$)	Bc (raw; prior $N(0, \Sigma_{ms})$)
3a	c	W_+Bc (pseudo-inverse; prior $N(0, V_+V_+^\top)$)
3b	(c, ζ), $\zeta \in \mathbb{R}^6$	$W_+Bc + V_0\zeta$ (exact $N(0, I)$)
4	θ; w.p. $\frac{1}{2}$ each: $c_g = \Pi_g\theta$, pCN step, $\theta' = B_g c_g'$	primal or dual grid
5	c: $K = 5 \cdot 10^4$ primal iterations, then $c = \Pi_{\text{dual}}\theta$ and $5 \cdot 10^4$ dual	raw, $\theta = B_g c$

Primal grid: 4×4 blocks, 4 local modes each. Dual grid: 5×5 tiling with interfaces at $(i + \frac{1}{2})/4$, block dimensions from the 95% local-energy rule ($p = 114$). Runs: 64 chains, $N = 10^5$, $K = 5 \cdot 10^4$, identical seeds across experiments; θ and local coefficients saved every 100 iterations, relative error and acceptance every iteration. About 103 s per chain.

3 Tests before sampling

3.1 Σ_{ms} rank test

grid	p	r_+ (of 64)	$d_{\min}$	cond(Σ_{ms})	result
primal	64	58	-2.9×10^{-16}	∞	FAIL
dual	114	64	0.284	10.7	PASS

The primal B (64×64) has a 12-orders-of-magnitude gap in its singular values after the 58th (Fig. 1); an independent MATLAB computation of $G = B^\top B$ agreed (deficient by exactly six). By decision the study proceeds with four modes per block and splits experiment 3 into 3a/3b. Whitening with the null-space term has operator error 6×10^{-15}.

3.2 Prior test (likelihood off)

With $\Phi \equiv 0$ every proposal is accepted; the chains should reproduce $N(0, I_{64})$ if the sampler preserves the prior. 64 chains, 4×10^4 iterations, $s = 0.5$.

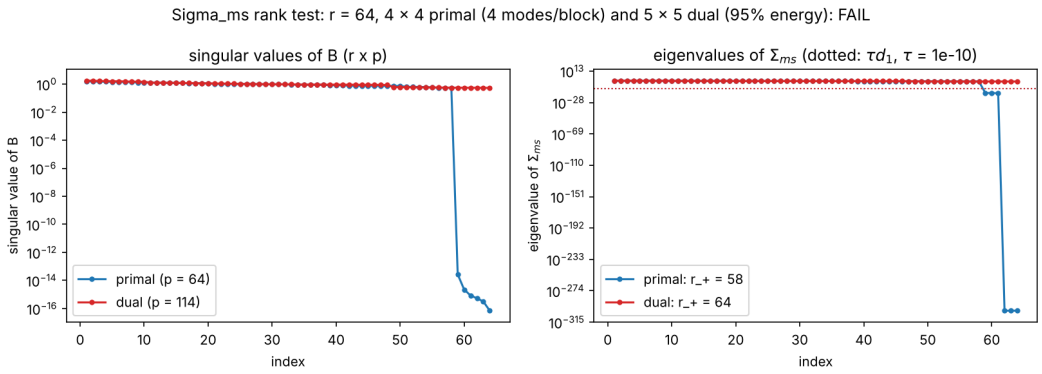


Figure 1: Singular values of B and eigenvalues of Σ_{ms}, primal and dual.

sampler	cov. error (Frob/$\sqrt{r}$)	pointwise var. ratio (mean)	range
reference	0.022	1.000	0.99–1.01
1 global	0.023	1.000	0.99–1.01
2 primal raw	0.631	0.783	0.31–1.22
3a pseudo-inverse	0.307	0.925	0.80–1.01
3b null-space term	0.023	1.001	0.99–1.01
4 mixture	0.422	0.619	0.51–0.97
5 phase 1 / phase 2	0.631 / 0.741	0.783 / 0.752	0.31–1.21 / 0.35–1.32

3a's error 0.307 matches the exact value $\sqrt{6/64} = 0.306$ for dropping six unit-variance directions. Alternating primal and dual moves (experiment 4) does *not* remove the pointwise variance deficit; it deepens it. The switch in experiment 5 changes the field by 8% (relative L_2) on prior draws.

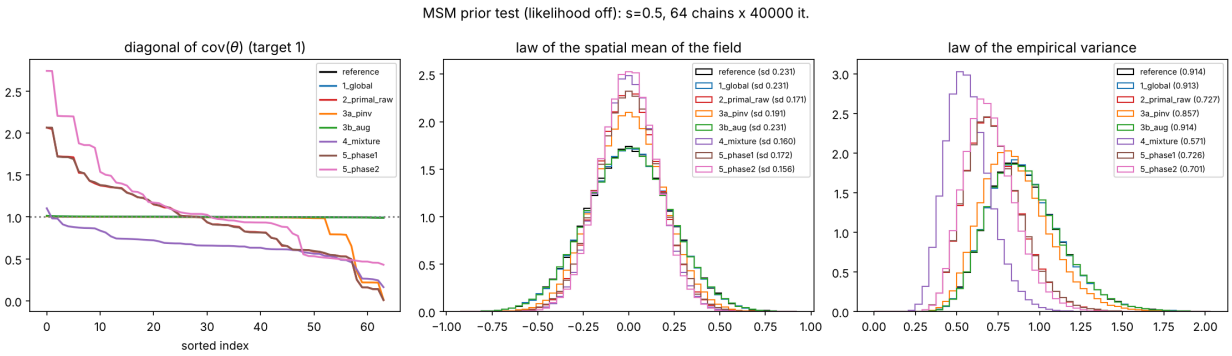


Figure 2: Prior test: variances of θ_i, law of the spatial mean, law of the empirical field variance.

4 Production results

4.1 Adaptation, misfit and mixing

	final s	mean AR	median relerr	MPSRF(θ)	MPSRF(c)	< 1.2 at it.
1	0.041	0.254	0.0205	1.127	—	72 023
2	0.051	0.253	0.0204	1.078*	1.096	44 473
3a	0.041	0.253	0.0204	1.094*	1.109	59 144
3b	0.041	0.262	0.0205	1.120	1.138†	68 506
4	0.001 / 0.041	0.002 / 0.249	0.0332	1.611	1.568 / 1.828	never
5 ph. 1	0.051	0.253	0.0204	1.169*	1.205	44 473
5 ph. 2	0.051–0.064	0.265	0.0207	1.165	1.297	90 958
5 full run				1.075		44 432

* in the 58 range coordinates $V_+^\top \theta$ (the θ of these runs lies in a 58-dimensional subspace, so the 64-dimensional within-chain covariance is singular); † joint (c, ζ), 70 dimensions. Experiment 4: primal / dual; MPSRF of the two projections $\Pi_p\theta$ / $\Pi_d\theta$. MPSRF uses the MATLAB-convention curve of `brho/mpsrf.py` (skip 2, step 15, burn 10) on the 900 saved samples per chain after M (400 per phase for 5); the last column is the first crossing of 1.2.

Acceptance and step sizes. Adaptation reaches the 25% target everywhere except the primal move of experiment 4 (Fig. 4). **Misfit.** Every healthy sampler has median relerr ≈ 0.0204 after adaptation (traces shown for 1 and 5 in Fig. 3), i.e. $\Phi \approx \frac{1}{2}(0.0204/0.003)^2 \approx 23$ (measured median 23.6 before the switch in 5). This is $\approx 6.4\times$ the data noise (0.0032) and is the expected posterior level for this problem; the traces show no drift after M. **Mixing.** θ-MPSRF is between 1.08 and 1.13 for 1–3b (the global control is highest, 1.127), all below 1.2; experiment 5's phase-wise values (1.17, 1.17 from 400 samples each) and the local coefficients of phase 2 (1.30) are less settled simply because the windows are shorter.

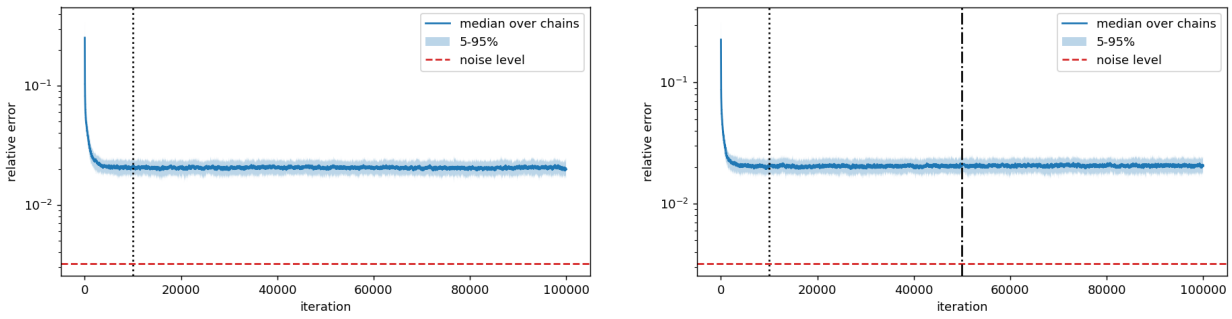


Figure 3: Relative error, median and 5–95% band over 64 chains. Left: experiment 1. Right: experiment 5 (dotted: end of adaptation; dash-dotted: primal→dual switch). Dashed red: realised noise level.

4.2 Experiment 5: the primal→dual switch

Projecting each chain's final global field onto the dual basis changes Φ from 23.57 to 24.80 (median over chains) and relerr from 0.0206 to 0.0211: a small step that is not visible at the scale of Fig. 3. Both phases then run at the 25% acceptance level with unchanged relative error.

4.3 Experiment 4: why it fails

The primal move proposes $\theta' = B_p\left(\sqrt{1-s^2}\,\Pi_p\theta + s\xi\right)$. Even at $s = 0$ this is the lossy round trip $\theta \mapsto B_p\Pi_p\theta$, which changes the field by 19% (relative L_2, prior draws; dual: 5.7%). Against a likelihood this sharp (the chains sit at relerr $= 0.02$ and $\sigma_{rel} = 0.003$), such proposals are almost always rejected: the adapter drives s to its floor 10^{-3} without lifting acceptance (mean 0.16%, Fig. 4). Effectively only the dual move works, with $\approx$ half the iterations wasted and a round trip that is itself lossy. The chains plateau at relerr $= 0.033$, MPSRF is 1.6 (no crossing of 1.2), and the pointwise sd of $\log\kappa$ is 0.22, so the result is a non-converged, biased sampler rather than a posterior. The mechanism (round-trip field change) is inferred from prior draws and the acceptance data, not measured at posterior states.

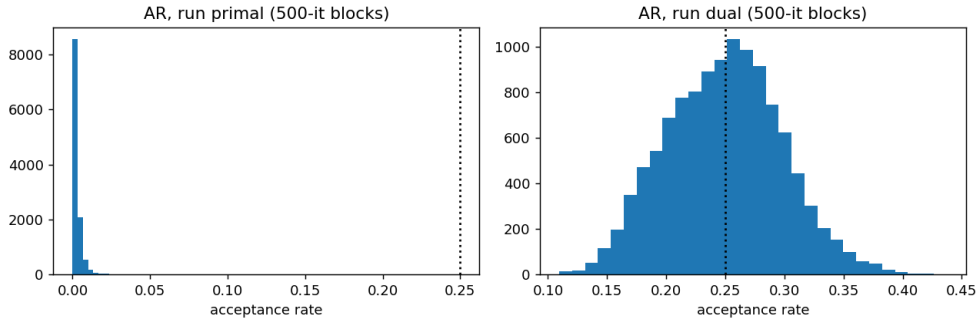


Figure 4: Experiment 4 acceptance rate (500-iteration blocks, 64 chains) for the primal (left) and dual (right) moves.

4.4 Posterior fields

Pointwise mean and sd from 5 000 post-M samples per experiment (2 500 per phase for 5).

	mean sd $\log\kappa$	ratio to exp. 1	mean sd p	rms(mean $\log\kappa$−truth)
1 global	0.570	1.00	0.0094	0.415
2 primal raw	0.473	0.83	0.0091	0.495
3a pseudo-inverse	0.511	0.90	0.0092	0.456
3b null-space term	0.575	1.01	0.0093	0.411
4 mixture	0.217	0.38	0.0063	0.556
5 (phase 1 / 2 / pooled)	0.468 / 0.472 / 0.492	0.82 / 0.83 / 0.86	0.0091 / 0.0094 / 0.0094	0.495 / 0.437 / 0.445

The sd of $\log\kappa$ orders the samplers as the prior test predicts: $3b \approx 1 > 3a > 5 \approx 2 \gg 4$. The pressure sd, which is constrained by the data, differs by about 3% at most among 1–3b and 5. The posterior mean of $\log\kappa$ is closest to the truth for 3b and 1 (rms 0.41) and farthest for 2 and 4. The raw samplers' narrower posteriors do not buy a better mean, they also fit the truth worse.

5 Interpretation

- The six-direction rank deficit of the primal Σ_{ms} is the whole difference between 3a and 3b: with the null-space term the whitened primal sampler is exactly global pCN; without it the sampler cannot move along six directions and the posterior is slightly too narrow (sd 0.90×, and the prior test gives 0.93 in variance).

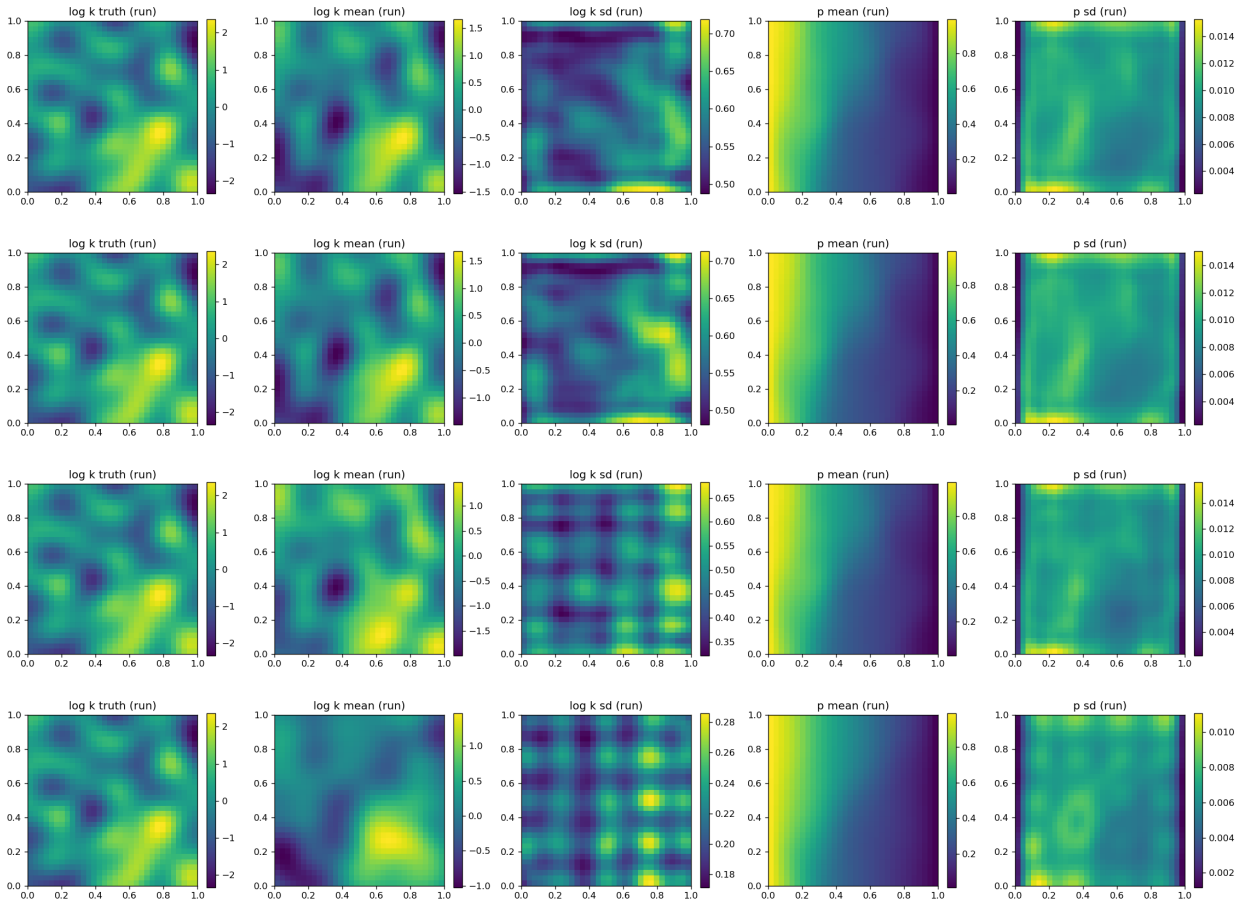


Figure 5: Truth, posterior mean and sd of $\log \kappa$, posterior mean and sd of the pressure, for experiments 1, 3b, 2 and 4 (top to bottom). Each panel has its own color scale. These are the first-run figures (default colormap); the group-style contour plots are regenerated by `run_msm_post.py` (Section 7).

- Raw primal sampling (2) targets the wrong prior, $N(0, \Sigma_{ms})$ whose per-coordinate prior variances range from 0.01 to 2.1 (prior test). It is nevertheless as fast as the others per iteration and acceptance, so it gains little in mixing (MPSRF 1.08 vs. 1.13; no uncertainty on that difference is available), and the price is 17% less posterior spread and a worse posterior mean.
- Experiment 5 inherits raw primal's variance deficit (and the dual's, 0.75 in the prior test) but loses little at the switch; the full-run θ MPSRF of 1.075 includes both phases.
- Experiment 4 shows that alternating with a lossy projection of a *state* is not a substitute for proposing in the local coefficients (experiments 2 and 5): the projection error enters every proposal.

6 Caveats

- One data set and one set of chain seeds. The 10–17% differences in posterior sd (2, 3a, 5 vs. 1) are consistent across the prior test and production but have no repeated-data or bootstrap uncertainty here.

- MPSRF is 1.08–1.14, not below 1.05; sd estimates from 64 chains of 900 saved samples are themselves uncertain by an amount not quantified here. For 2, 3a and phase 1 of 5 it is computed in range coordinates, so values are not directly comparable in dimension (58 vs. 64).
- The likelihood is a global-norm relative likelihood with $\sigma_{rel} = 0.003$ and data noise equal to it; the plateau at relerr = 0.020 (about 6.4× the noise) means the posterior is dominated by what the 64 parameters and the likelihood weight allow, not by the noise floor. With a tempered likelihood (as in the repo's earlier runs, $\sigma_{rel} = \varepsilon/\sqrt{n_{\text{eff}}}$) the picture could differ.
- Experiment 4's failure mechanism is inferred, not measured at posterior states.
- Field figures here are not in the final plotting style (below).

7 Reproduction and next steps

Code: `brho/msm_local.py`, `brho/msm_sampler.py`, `experiments/run_msm_chains.py`, `run_msm_post.py`, `run_msm_compare.py`; six `cluster/slurm/msm_exp*.sbatch`; spec `docs/msm_local_kle_experiment.md`; outputs `runs/msm_local_kle_2026-11/`.

Field plots. `experiments/msm_fieldplots.py` draws filled contours with `jet` (colormaps for fields and sd maps are separate defaults, `CMAP_MAIN`, `CMAP_STD`, 20 levels; `--cmap`, `--cmap-std`, `--levels` on the command line). The posterior field arrays are now saved to `post/fields.npz` and kept in git, so after one rerun of `run_msm_post.py` on the cluster per experiment, `--replot-only` redraws any style without the chains, and `run_msm_compare.py` produces six-panel sd/mean maps on shared scales and a table.

Suggested next steps. (i) Regenerate the fields in the chosen style and add the six-panel comparison; (ii) repeat with a second truth seed to put an error bar on the sd ratios; (iii) if experiment 4 is to be kept, propose in the local coefficients of a persistent state, or restrict it to the dual grid; (iv) decide whether the primal grid should use 6 local modes per block, which makes Σ_{ms} full rank (as a separate study).